

AEGIS-TWIN

Forensic Incident Report

SEVERITY: CRITICAL

Device Name	Entrance IP Security Camera	Device ID	RPI-IPCAM-01
Director / Zone	IoT Security	Timestamp	10 May 2026 02:29:46 UTC
Trust Score	0.0 / 100	Reconstruction MSE	0.99960
Entropy-Shannon Div	0.31290	Attack Pattern	Undetermined — insufficient distinguishing info

Incident Signature (SHA-256)	B95C46557E121AB18FD12284159C74E95AD8D7B2E7817440B2A2CCF4D2E580D8
---------------------------------	--

1 — Executive Summary

A **CRITICAL** severity anomaly was detected. The device exhibits behaviour strongly inconsistent with its established baseline and must be treated as actively compromised. Immediate isolation and incident response are required.

The Aegis-Twin digital twin engine, using an LSTM autoencoder, observed device **Entrance IP Security Camera** (ID: **RPI-IPCAM-01**) in sector **IoT Security** at **10 May 2026 02:29:46 UTC**. The trust score fell to **0.0 / 100** and reconstruction error rose to **0.99960**, triggering automated forensic collection and report generation.

2 — Digital Twin Analysis Metrics

The following metrics were recorded by the autoencoder at the time of detection. Elevated MSE and JSD values indicate that the observed traffic distribution diverges significantly from the trained baseline.

Metric	Observed Value	Reference / Threshold
Trust Score	0.0 / 100	Healthy range: ≥ 75
Reconstruction Error (MSE)	0.999600	Threshold: 0.20 (medium) / 0.35 (high) / 0.50 (critical)
Jensen-Shannon Divergence	0.312900	Threshold: 0.30 / 0.50 / 0.70
Derived Severity	CRITICAL	Computed from the above three metrics

3 — Top Anomaly Contributing Features

The features below showed the largest deviation between the observed traffic and the trained baseline. Higher delta values indicate greater anomaly contribution.

Feature	Baseline	Observed	Delta	% Change	Direction
Packet Size	0.1200	0.5970	0.4770	397.5%	↑ Increased
Inter-Arrival Time (IAT)	0.3500	0.1131	0.2369	67.7%	↓ Decreased
Payload Entropy	0.1800	0.3129	0.1329	73.8%	↑ Increased

4 — Digital Evidence Snapshot

Raw feature vectors captured at detection time for archival and chain-of-custody purposes.

Feature	Baseline	Current	Abs. Delta
Packet Size	0.12000	0.59700	0.47700
Inter-Arrival Time (IAT)	0.35000	0.11310	0.23690
Payload Entropy	0.18000	0.31290	0.13290
Flow Symmetry	0.52000	0.52630	0.00630

■ 5 — Identified Attack Pattern

Classified pattern: Undetermined — insufficient distinguishing indicators

The classification is derived from heuristic analysis of feature deltas and packet inter-arrival timing distributions. It should be treated as an investigative hypothesis and validated by a human analyst.

■ 6 — Packet Traffic Summary

No packet history records were present at the time of report generation.

■ 7 — Behavioural Timeline

Timestamp	Level	Event Description
2026-05-10T02:29:48.486849+00:00	INFO	SYN Flood detected on IP Camera — trust dropped to 0.0. pkt_size=0.597 iat=0.113 entropy=0.313 symmetry=0.526

■ 8 — Risk Assessment

This incident is rated **CRITICAL**. The combination of very low trust score, high reconstruction error, and significant distribution divergence indicates a strong likelihood of active exploitation or data exfiltration. The device should be treated as **fully compromised** until cleared by forensic investigation.

■ 9 — Recommended Remediation Actions

Priority	Remediation Action
Immediate	Isolate the device from the network (quarantine VLAN or physical disconnect).
Immediate	Preserve volatile memory: capture a RAM image before any reboot.
Short-term	Collect full packet captures and system/application logs for post-incident analysis.
Short-term	Perform a firmware integrity check and compare against known-good hash.
Short-term	Rotate all credentials, API keys, and revoke active sessions associated with the device.
Long-term	Update IDS/IPS signatures to detect similar behavioural patterns.
Long-term	Review and tighten network segmentation and access-control lists for this sector.
Long-term	Schedule a post-incident review and update the Aegis-Twin baseline after remediation.

CONFIDENTIAL — FOR AUTHORIZED PERSONNEL ONLY | This report was generated automatically by the Aegis-Twin Forensic Engine. All findings should be reviewed by a qualified security analyst before action is taken.